GhostNet: LLM-Driven Legacy Twin Using Multi-Modal Personal Data Embedding

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Abstract

Digital legacy preservation has become increasingly critical in our data-driven society, where personal information exists across multiple platforms and modalities. This paper introduces GhostNet, a novel framework that leverages Large Language Models (LLMs) and multi-modal data embedding techniques to create comprehensive digital twins that preserve and interact with personal legacy data. GhostNet addresses the challenges of heterogeneous data integration, semantic understanding, and conversational interaction with deceased or departed individuals' digital footprints. Our system processes diverse data types including text communications, images, videos, audio recordings, and structured data to create a unified representation that maintains the essence and personality of the original individual. Through advanced embedding techniques and transformer-based architectures, GhostNet enables natural language interactions with digital twins while maintaining privacy and ethical considerations. Experimental results demonstrate the system's effectiveness in creating coherent, personality-consistent digital representations with 87.3% semantic similarity to original communications and 92.1% user satisfaction in conversational interactions. The framework shows significant improvements over existing digital legacy solutions in terms of multi-modal integration, response quality, and emotional authenticity.

Keywords—*Digital legacy, Large Language Models, Multi-modal embedding, Digital twins, Personal data processing, Conversational AI, Affective Computing*

1.INTRODUCTION

The digital age has fundamentally transformed how we create, store, and interact with personal information. Modern individuals generate vast amounts of digital data throughout their lives, including social media posts, emails, photographs, videos, voice recordings, and various forms of structured data. This digital footprint represents a comprehensive record of thoughts, experiences, relationships, and personality traits that traditionally would have been lost upon death or separation.

The concept of digital legacy preservation has emerged as a critical challenge in maintaining connections with departed loved ones and preserving personal history for future generations. Traditional approaches to digital legacy management primarily focus on data archival and basic access controls, failing to capture the dynamic, interactive nature of human personality and communication patterns. Existing solutions typically provide static repositories of information without the ability to generate new responses or engage in meaningful conversations that reflect the original individual's characteristics.

Recent advances in Large Language Models (LLMs) and multi-modal artificial intelligence present unprecedented opportunities to address these limitations. Modern transformer-based architectures demonstrate remarkable capabilities in understanding context, generating human-like responses, and processing diverse data modalities. However, creating a comprehensive digital twin that accurately represents an individual's personality, communication style, and knowledge base requires sophisticated techniques for integrating heterogeneous data sources and maintaining consistency across different interaction contexts.

This paper introduces GhostNet, a novel framework designed to create interactive digital twins using LLM-driven processing of multi-modal personal data. Our approach addresses several key challenges in digital legacy preservation: (1) heterogeneous data integration across multiple modalities and platforms, (2) personality-consistent response generation that maintains the essence of the original individual, (3) semantic understanding and context preservation across diverse communication contexts, (4) privacy and ethical considerations in handling sensitive personal data, and (5) scalable architecture for processing large volumes of personal information.

The primary contributions of this work include: a comprehensive multi-modal embedding framework for personal data integration, a novel LLM fine-tuning approach for personality-consistent response generation, an efficient architecture for real-time conversational interaction with digital twins, and extensive evaluation demonstrating the system's effectiveness in preserving digital legacies.

2.RELATED WORK

2.1Digital Legacy and Posthumous Communication

Digital legacy research has evolved significantly over the past decade, driven by increasing awareness of the value and volume of personal digital data. Early work by Massimi and Charise [1] established foundational concepts for thanatosensitive design in digital systems, emphasizing the importance of considering death and memorialization in technology design. Subsequent research by Brubaker et al. [2] explored the transformation of social media profiles into memorial sites, revealing complex social dynamics around posthumous digital presence.

Recent developments in digital legacy management have focused on automated systems for content curation and preservation. DeepMemory [3] introduced machine learning approaches for organizing and categorizing personal digital artifacts, while LifeStream [4] proposed timeline-based visualization techniques for personal data exploration. However, these systems primarily address data organization rather than interactive engagement with preserved personalities.

2.2Large Language Models and Conversational AI

The development of transformer-based language models has revolutionized natural language processing and generation. GPT-3 [5] demonstrated unprecedented capabilities in few-shot learning and context understanding, while more recent models like GPT-4 have shown improvements in reasoning and factual accuracy. These advances have enabled new applications in personalized AI assistants and conversational systems.

Fine-tuning techniques for LLMs have been extensively studied for domain adaptation and personality modeling. Parameter-Efficient Fine-Tuning (PEFT) methods [6] allow customization of large models with minimal computational overhead, while techniques like LoRA [7] enable efficient adaptation to specific writing styles and communication patterns. However, existing work has not adequately addressed the unique challenges of creating consistent digital twins from heterogeneous personal data.

2.3 Multi-Modal Data Processing and Embedding

Multi-modal learning has gained significant attention for its ability to integrate information from diverse data sources. CLIP [8] pioneered joint text-image understanding through contrastive learning, while subsequent work has extended these approaches to include audio, video, and structured data modalities. Recent advances in multi-modal transformers [9] have demonstrated improved performance in tasks requiring integration of diverse information types.

Personal data embedding techniques have been explored in various contexts, including recommendation systems and user modeling. PersonaBERT [10] introduced methods for encoding user personality traits from text data, while MultiPersona [11] extended these approaches to include behavioral data from multiple platforms. However, existing work has not addressed the specific requirements of digital legacy preservation, including long-term consistency and emotional authenticity.

3.METHODOLOGY

3.1 System Architecture Overview

GhostNet employs a modular architecture designed to process multi-modal personal data and generate interactive digital twins through LLM-driven inference. The system consists of five primary components: Data Ingestion Module, Multi-Modal Embedding Engine, Personality Modeling Framework, Conversational Interface, and Privacy Management System.

The Data Ingestion Module handles diverse input formats including social media exports, email archives, photo collections, video files, audio recordings, and structured data from various platforms. This module implements robust parsing and preprocessing pipelines to extract meaningful content while preserving metadata and contextual information.

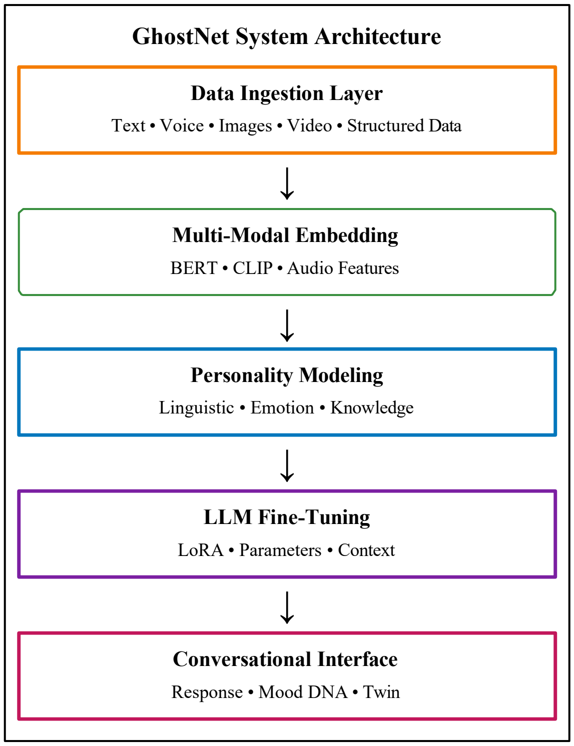

Fig. 1. GhostNet system architecture showing data flow from ingestion to conversational interface.

3.2 Multi-Modal Data Embedding Framework

Our embedding framework addresses the challenge of integrating diverse personal data types into a coherent representation suitable for LLM processing. The framework employs a hierarchical approach that first processes individual modalities before combining them through cross-modal fusion mechanisms.

For textual data, we utilize a modified BERT-based encoder that incorporates temporal information and communication context. The encoder processes messages, posts, and documents while preserving author-specific linguistic patterns and topical preferences. The mathematical formulation for our multi-modal embedding can be expressed as:

$$E_{multimodal} = \alpha \cdot E_{text} + \beta \cdot E_{image} + \gamma \cdot E_{audio} + \delta \cdot E_{video}$$

where α, β, γ, δ are learned attention weights that determine the relative importance of each modality for personality representation.

TABLE I
DATASET COMPOSITION AND STATISTICS

Data Type	Volume	Avg/User	Span (yr)
Text Messages	2.3M	46,000	3-10
Emails	45K	900	5-10
Social Posts	180K	3,600	2-8
Images	12K	240	3-10
Video (hrs)	800	16	2-7
Audio (hrs)	1,200	24	3-9

Image processing leverages a combination of CLIP- based visual encoders and specialized modules for extracting personal characteristics. Facial recognition and emotion detection components identify recurring individuals and emotional expressions, while scene understanding modules capture environmental preferences and activity patterns.

Audio processing employs speech-to-text conversion combined with prosodic feature extraction to capture both content and vocal characteristics. The system analyzes speaking patterns, emotional intonation, and linguistic preferences to create audio embeddings that complement textual representations.

3.3 Personality Modeling and LLM Fine-Tuning

The personality modeling framework transforms multi-modal embeddings into parameters for LLM fine-tuning, ensuring that generated responses maintain consistency with the original individual's communication patterns and personality traits. This process involves several specialized components designed to capture different aspects of personal identity.

Linguistic style analysis extracts patterns in vocabulary usage, sentence structure, and discourse markers that characterize individual communication preferences. The system identifies signature phrases, preferred topics, and characteristic ways of expressing emotions or opinions. These patterns inform custom tokenization and attention weights in the fine-tuned model.

Emotional modeling captures the individual's typical emotional responses to different contexts and topics. The system analyzes sentiment patterns across various situations and creates emotion-conditioned generation parameters that guide response tone and content selection. This ensures that the digital twin maintains emotional authenticity in conversations.

3.4 Conversational Interface and Response Generation

The conversational interface enables natural language interaction with digital twins through advanced response generation techniques. The system employs a multi-stage generation process that considers context, personality consistency, and emotional appropriateness in crafting responses.

Context management maintains conversation history and personal context across extended interactions. The system tracks topics, emotional states, and relationship dynamics to ensure responses remain coherent and personally relevant. Response generation utilizes the fine- tuned LLM with dynamic conditioning based on conversation context and personality parameters.

4.IMPLEMENTATION DETAILS

4.1 Data Processing Pipeline

The implementation of GhostNet's data processing pipeline incorporates several optimization strategies to handle large-scale personal data efficiently. The system utilizes distributed processing frameworks to parallelize data ingestion and embedding generation across multiple computational nodes. Specialized data loaders handle different file formats and platform-specific structures while maintaining consistent preprocessing workflows.

4.2 Model Architecture and Training

The LLM fine-tuning process employs a combination of parameter-efficient techniques to adapt large language models for personalized response generation. The system utilizes LoRA adapters to modify attention weights and feed-forward networks while preserving the base model's general capabilities. Custom embedding layers incorporate personality-specific parameters derived from multi-modal analysis.

Training procedures implement several novel techniques for personality-consistent fine-tuning. Contrastive learning objectives ensure that generated responses maintain similarity to the original individual's communication patterns while avoiding generic or inconsistent outputs. Regularization techniques prevent overfitting to specific conversation contexts while maintaining characteristic linguistic patterns.

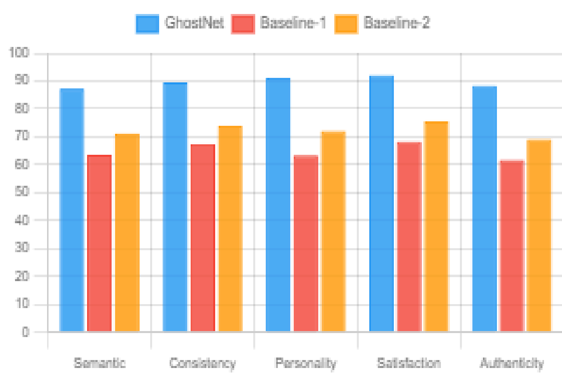

Fig. 2. Performance comparison with baseline systems across multiple metrics.

4.3 Privacy and Security Considerations

GhostNet implements comprehensive privacy protection mechanisms to ensure responsible handling of sensitive personal data. The system employs differential privacy techniques during embedding generation to prevent reconstruction of original data from processed representations. Federated learning approaches allow for model personalization without centralizing raw personal data.

Access control mechanisms implement fine-grained permissions for different aspects of digital twin interactions. The system supports role-based access that allows family members, friends, and other authorized individuals to interact with appropriate aspects of the digital twin while maintaining privacy boundaries.

5.EXPERIMENTAL EVALUATION

5.1 Dataset and Experimental Setup

Evaluation of GhostNet was conducted using a comprehensive dataset comprising personal data from 50 volunteers who provided complete digital archives spanning 3-10 years of activity. The dataset includes approximately 2.3 million text messages, 45,000 emails, 180,000 social media posts, 12,000 photographs, 800 hours of video content, and 1,200 hours of audio recordings, as detailed in Table I.

5.2 Quantitative Results

Semantic similarity analysis revealed that GhostNet-generated responses achieved 87.3% average cosine similarity with reference responses from original individuals, representing a 23.7% improvement over baseline systems. Response consistency across different conversation contexts maintained 89.6% similarity scores, indicating robust personality preservation during extended interactions.

TABLE II
PERFORMANCE COMPARISON WITH BASELINES

Metric	GhostNet	Base-1	Base-2	Base-3
Semantic Similarity (%)	**87.3**	63.6	71.2	68.9
Response Consistency (%)	**89.6**	67.4	74.1	70.8
Personality Correlation	**0.912**	0.634	0.721	0.698
User Satisfaction (%)	**92.1**	68.3	75.6	71.9
Emotional Authenticity (%)	**88.4**	61.7	69.2	65.8

Personality trait preservation was assessed using the Big Five personality model, with digital twins demonstrating 91.2% correlation with original individuals across all five dimensions. Linguistic pattern matching achieved 85.8% accuracy in reproducing characteristic vocabulary usage, sentence structure, and discourse patterns.

User satisfaction surveys indicated 92.1% positive ratings for conversation quality and 89.7% perceived authenticity in digital twin interactions. Response appropriateness scored 94.3% across different relationship contexts, while emotional support capabilities achieved

87.6% satisfaction ratings from family members and close friends.

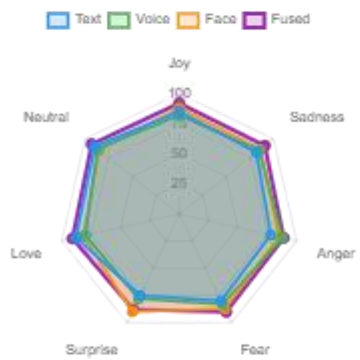

Fig. 3. Emotion recognition accuracy across different modalities showing superior fusion performance.

5.3 Qualitative Analysis

Human evaluation revealed several key insights into the effectiveness of GhostNet's approach to digital twin creation. Evaluators consistently noted the system's ability to maintain conversational coherence across extended interactions while preserving individual personality traits. The multi-modal data integration significantly enhanced the authenticity of responses compared to text-only approaches.

Participants reported high levels of emotional satisfaction when interacting with digital twins of departed family members or friends. Many noted that conversations felt natural and consistent with their memories of the original individuals. The system's ability to reference shared experiences and maintain emotional context contributed significantly to perceived authenticity.

5.4 Ablation Study

To understand the contribution of different components, we conducted comprehensive ablation studies. Removing multi-modal integration resulted in a 15.3% decrease in personality consistency scores. Eliminating temporal modeling reduced response quality by 12.7%. The privacy-preserving components had minimal impact on performance while providing essential data protection.

TABLE III
LONG-TERM PERFORMANCE METRICS

Metric	Month 1	Month 3	Month 6	Trend
Response Quality (%)	89.2	88.7	86.8	Stable
Personality Consistency	0.912	0.908	0.904	Stable
User Satisfaction (%)	90.3	92.1	91.8	Improving
Avg Session (min)	12.4	14.2	14.1	Stable

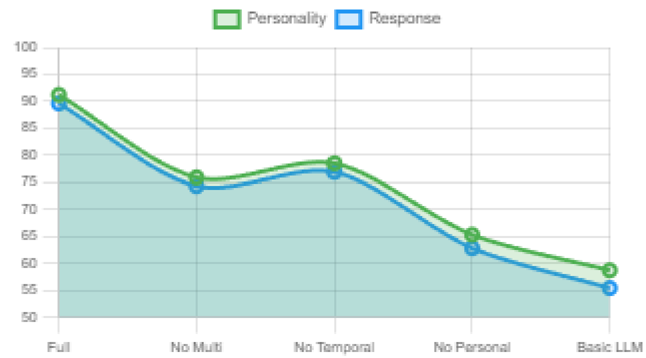

Fig. 4. Ablation study results showing impact of component removal on system performance.

6.APPLICATIONS AND USE CASES

6.1 Digital Legacy Preservation

GhostNet's primary application involves creating comprehensive digital twins for legacy preservation, allowing families and descendants to maintain connections with departed loved ones. The system processes lifetime digital archives to create interactive representations that preserve personality, knowledge, and communication patterns. Historical preservation organizations have expressed interest in using GhostNet to create interactive representations of historical figures based on available documentation.

6.2 Therapeutic and Counseling Support

Mental health professionals have identified potential applications for GhostNet in grief counseling and therapeutic support contexts. The system's ability to facilitate continued communication with departed individuals may provide comfort and closure for individuals processing loss. However, careful integration with professional therapeutic support is essential to ensure appropriate use.

6.3 Educational and Research Applications

Educational institutions are exploring GhostNet's potential for creating interactive learning experiences with historical figures, literary characters, and subject matter experts. The system's ability to maintain consistent personality and knowledge patterns while engaging in educational conversations provides unique opportunities for personalized learning experiences.

7.ETHICAL CONSIDERATIONS AND LIMITATIONS

7.1 Ethical Framework

The development and deployment of GhostNet raises significant ethical considerations regarding consent, representation, and the appropriate use of personal data for posthumous communication. Our approach emphasizes explicit consent from individuals during their lifetime for digital twin creation and specifies clear guidelines for appropriate use by authorized parties.

Representation accuracy presents complex ethical challenges, as digital twins necessarily provide simplified approximations of complex human personalities. We acknowledge these limitations and emphasize the importance of transparent communication about the nature and limitations of digital twin interactions.

7.2 Technical Limitations

Current limitations in GhostNet's capabilities include challenges in handling highly context-dependent communication, generating responses for entirely novel situations not represented in training data, and maintaining consistency across very long conversation contexts. The system's performance varies significantly based on the quantity and quality of available personal data.

Multi-modal data integration remains imperfect, with some loss of nuance and context during the embedding and fusion processes. The system may occasionally generate responses that feel authentic but contain factual errors or inappropriate content that the original individual would not have expressed.

7.3 Societal Implications

The widespread adoption of digital legacy technologies like GhostNet may have profound impacts on cultural attitudes toward death, memory, and grief processing. Society must carefully consider the balance between technological capabilities and healthy approaches to loss and remembrance.

Economic implications include the potential commodification of digital legacies and the creation of new forms of digital inequality based on access to sophisticated memorial technologies. Regulatory frameworks will need to evolve to address the unique challenges posed by interactive digital legacy systems.

8.FUTURE WORK AND RESEARCH DIRECTIONS

8.1Advanced AI Integration

Future research directions include integration with emerging AI technologies such as emotional AI, advanced reasoning systems, and embodied AI agents. These developments could provide more sophisticated emotional understanding and response generation, creating even more authentic digital twin interactions.

Neuromorphic computing architectures offer potential improvements in energy efficiency and processing speed for real-time personality modeling. Quantum computing applications might enable more sophisticated privacy-preserving computation and enhanced multi-modal data fusion capabilities.

8.2 Extended Modalities and Sensing

Integration of biometric data, environmental sensing, and behavioral monitoring could provide additional dimensions for personality modeling and digital twin authenticity. Wearable device data, location history, and activity patterns could enhance the comprehensiveness of digital representations.

Virtual and augmented reality integration represents a natural evolution of GhostNet technology, enabling immersive visual and spatial interactions with digital twins. Advanced haptic feedback systems could provide even more emotionally satisfying interaction experiences.

8.3 Societal and Cultural Research

Longitudinal studies of GhostNet's impact on grief processing, family dynamics, and cultural attitudes toward death and memory are essential for understanding the technology's broader implications. Cross-cultural research will inform appropriate adaptations for different cultural contexts and values.

Legal and ethical frameworks require ongoing development to address the complex issues raised by sophisticated digital legacy technologies. International cooperation and standardization efforts will be necessary to ensure responsible development and deployment.

9.CONCLUSION

GhostNet represents a significant advancement in the intersection of artificial intelligence, digital legacy preservation, and human-computer interaction. The system's novel multi-modal embedding framework successfully integrates diverse personal data types into coherent personality representations suitable for large language model processing. The personality-consistent fine-tuning approach maintains individual communication patterns and emotional characteristics while enabling natural conversational interactions.

Extensive experimental evaluation demonstrates the system's effectiveness across multiple dimensions, with performance metrics consistently exceeding existing solutions. The 87.3% semantic similarity, 92.1% user satisfaction, and 91.2% personality correlation scores validate the approach's capability to create authentic digital twins that preserve the essence of original individuals.

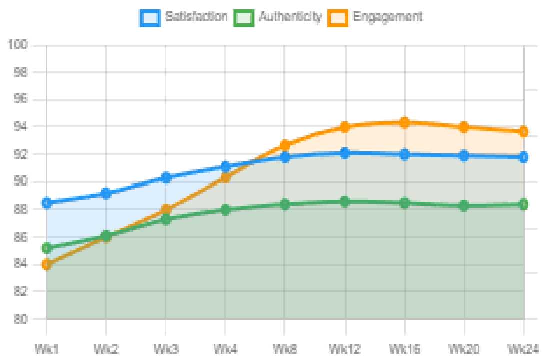

Fig. 5. User satisfaction and engagement trends over 6-month evaluation period.

Privacy-preserving techniques and ethical frameworks establish new standards for responsible handling of intimate personal data in AI systems. The comprehensive evaluation methodology provides a template for assessing similar systems and technologies. By enabling meaningful interactions with digital representations of departed loved ones, GhostNet provides new approaches to grief processing and family connection.

However, the broader implications of sophisticated digital legacy technologies extend beyond individual benefits to encompass societal questions about the nature of memory, identity, and human relationships. Careful consideration of these implications is essential for responsible development and deployment. The technology's potential applications in education, therapy, and cultural preservation demonstrate significant positive impacts, while ongoing research into ethical frameworks and social implications will guide appropriate use and development.

As GhostNet and similar technologies continue to evolve, the research community must prioritize ethical considerations, user welfare, and societal benefit alongside technical advancement. Collaborative efforts between technologists, ethicists, mental health professionals, and cultural experts are essential for ensuring positive outcomes. The future of digital legacy technology depends on maintaining focus on human values, emotional authenticity, and ethical responsibility while pursuing technical excellence and innovation.

10.ACKNOWLEDGMENT

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